

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 1
timesteps_orig: 30
timesteps: 30
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
forecast_steps: 5

LSTM Multi-Step Training Report

Report Generated: 2025-05-09 15:12:15

Training Start: 2025-05-09_13-23-53

Training End: 2025-05-09_14-15-30

Training Duration: 51.62 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 0.9145

MSE: 1.3388

RMSE: 1.1571

R2 Score: 0.9262

Multi-Step Forecast:

Forecast Steps: 5

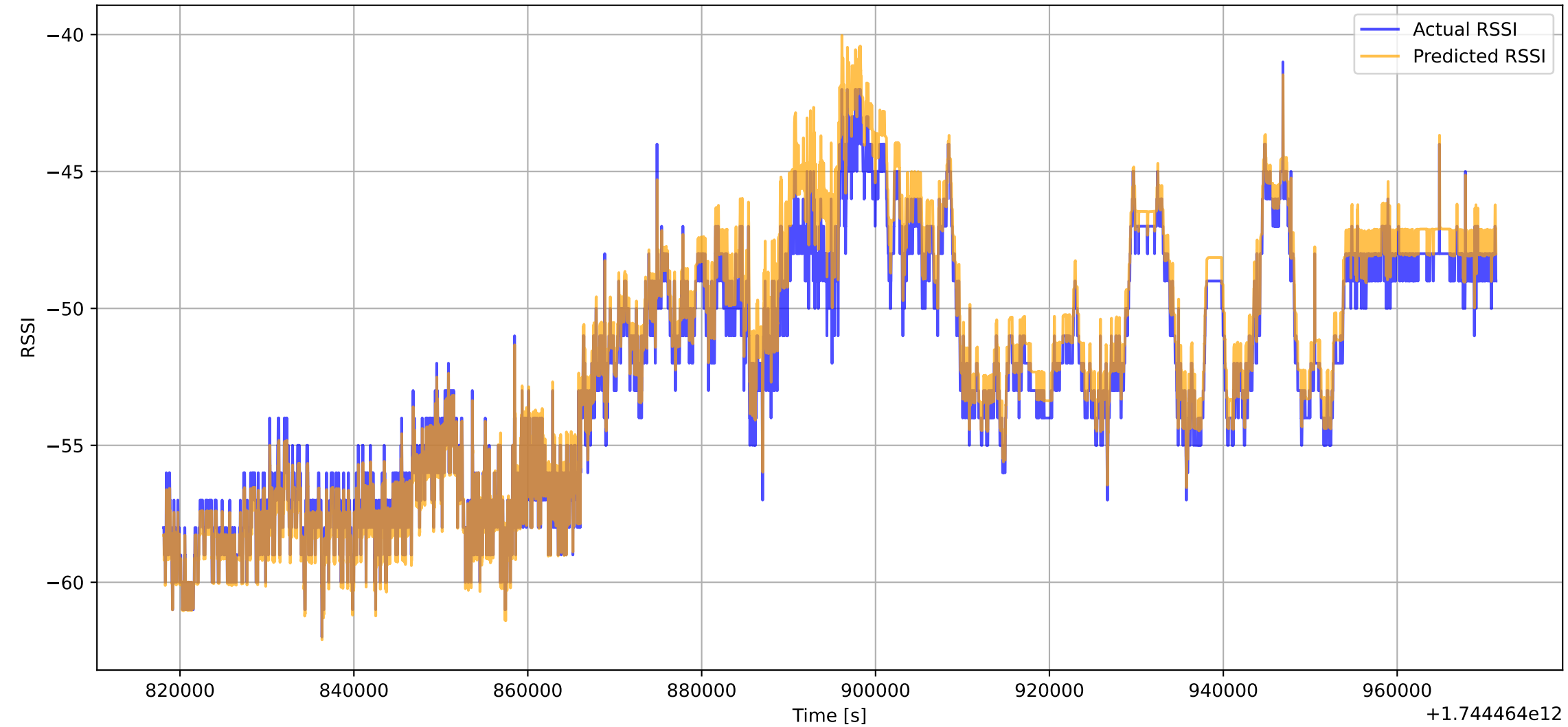
MAE: 0.0121

MSE: 0.0002

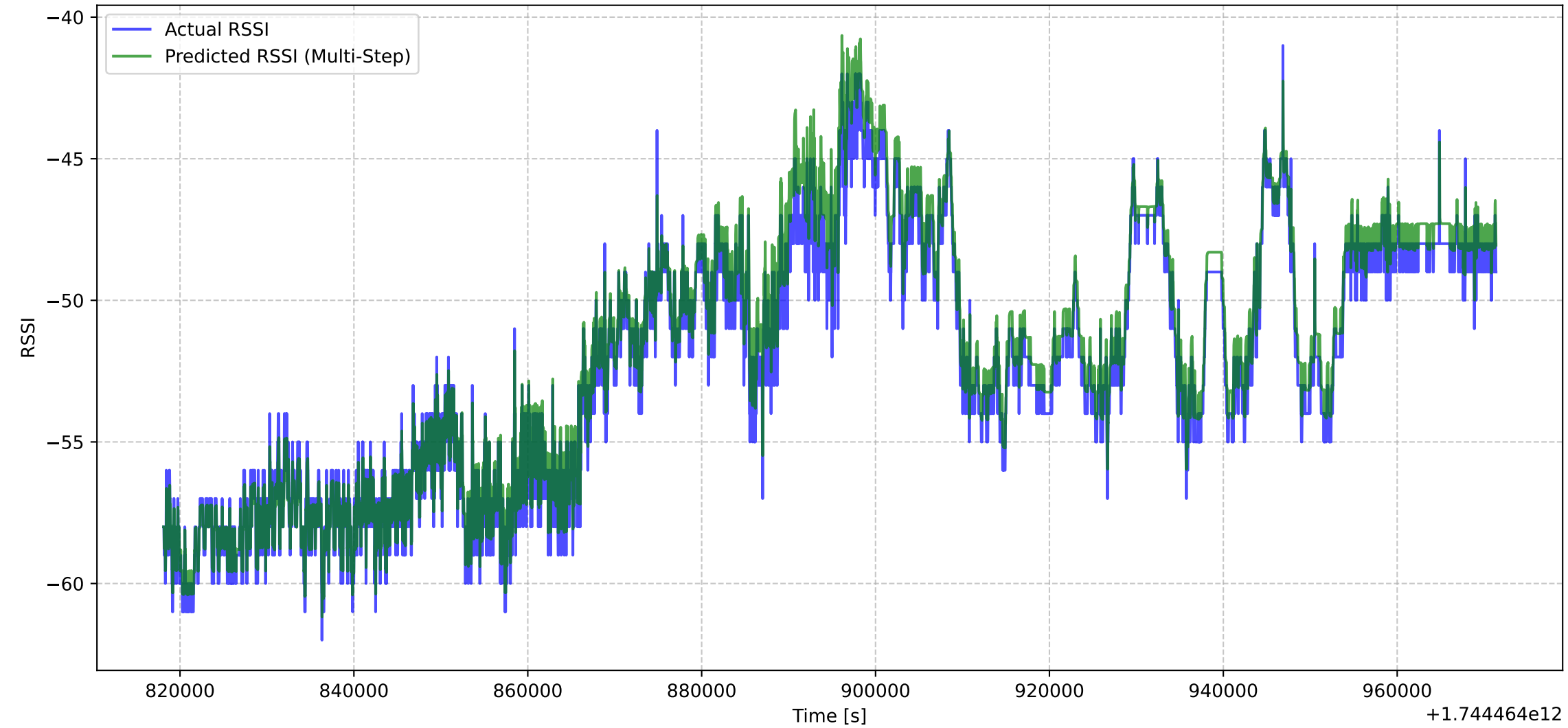
RMSE: 0.0151

R2 Score: 0.9437

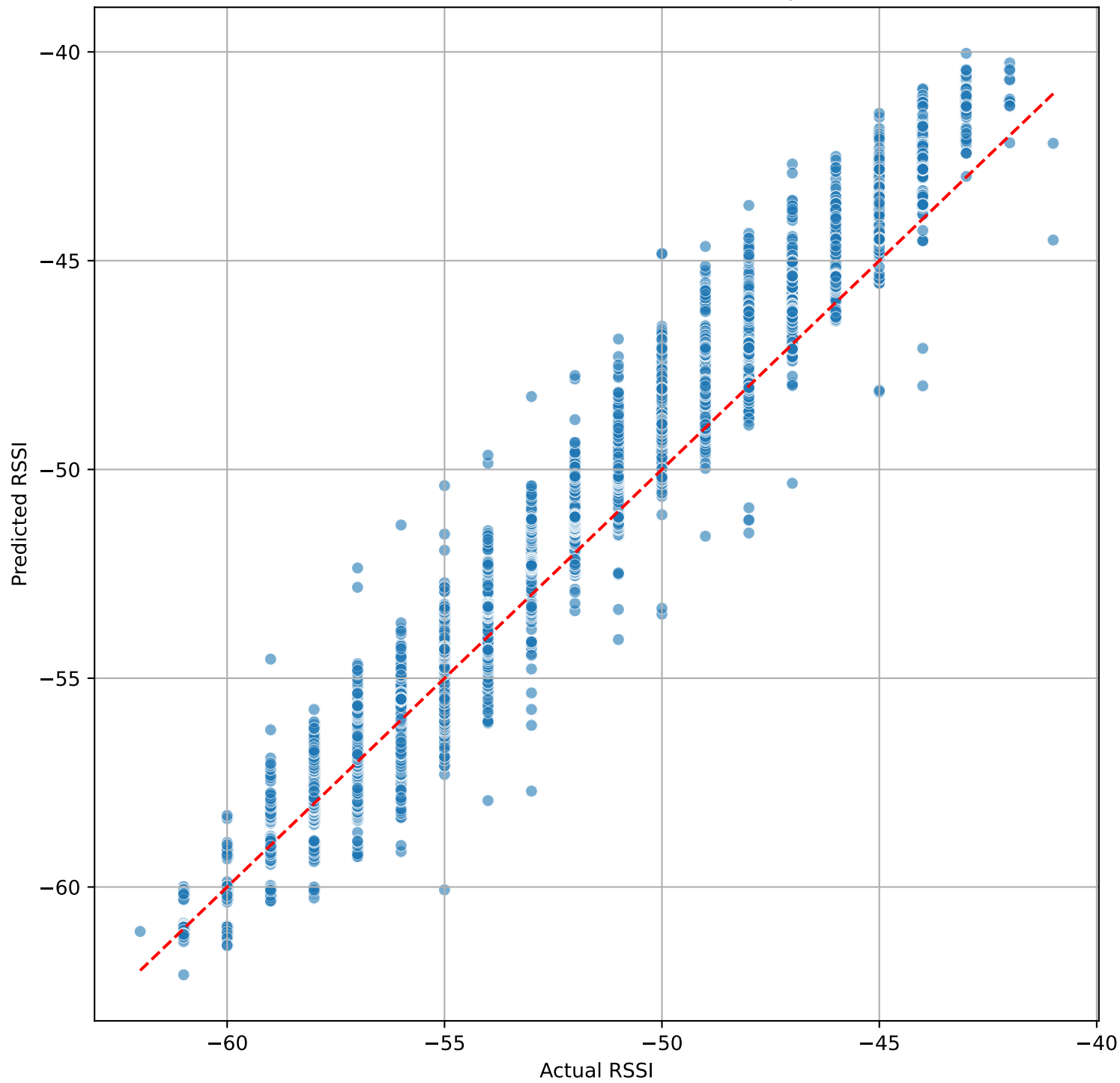
Actual vs Predicted RSSI (First Step)



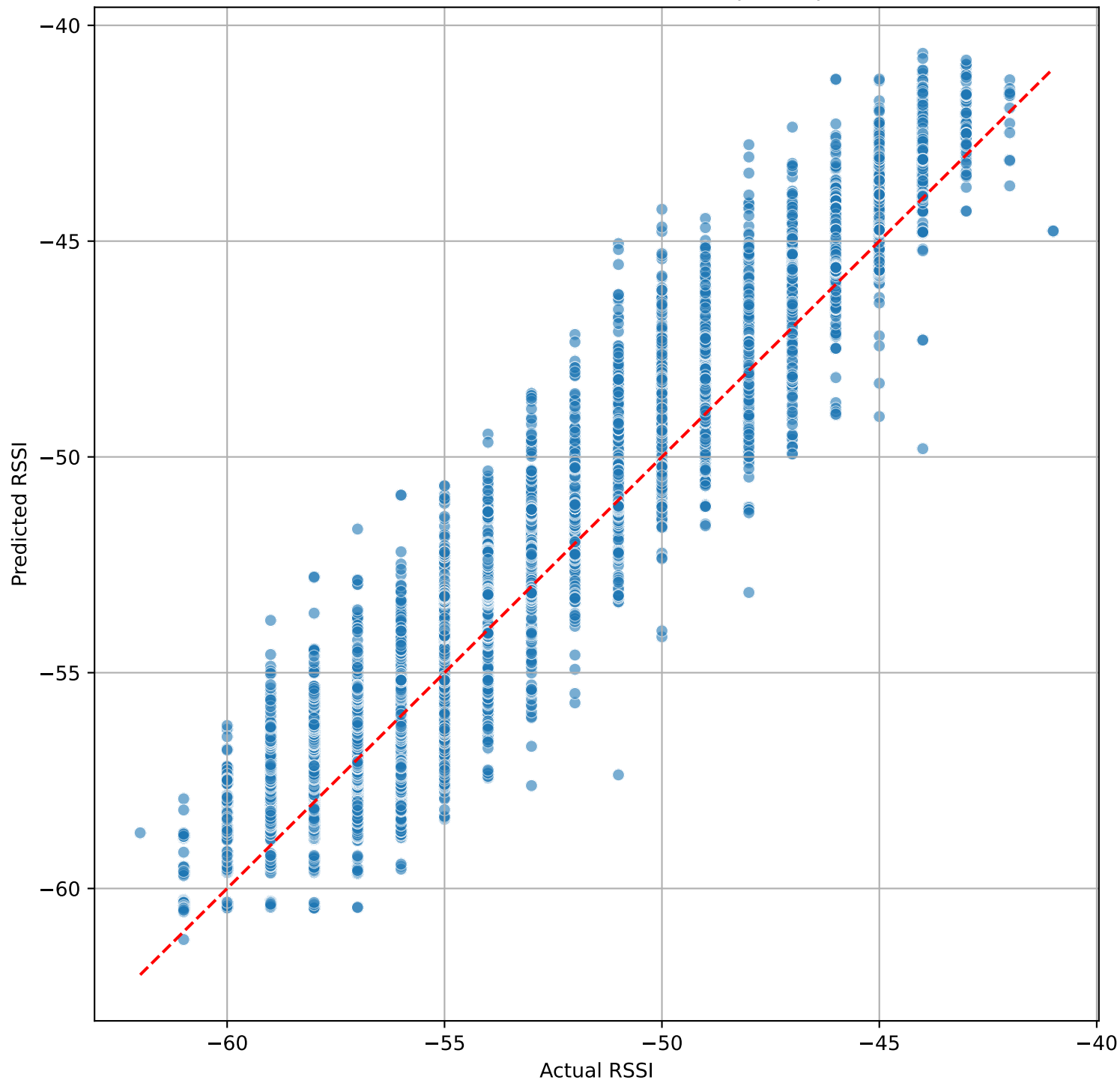
Actual vs Predicted RSSI (Multi-Step (step 5))



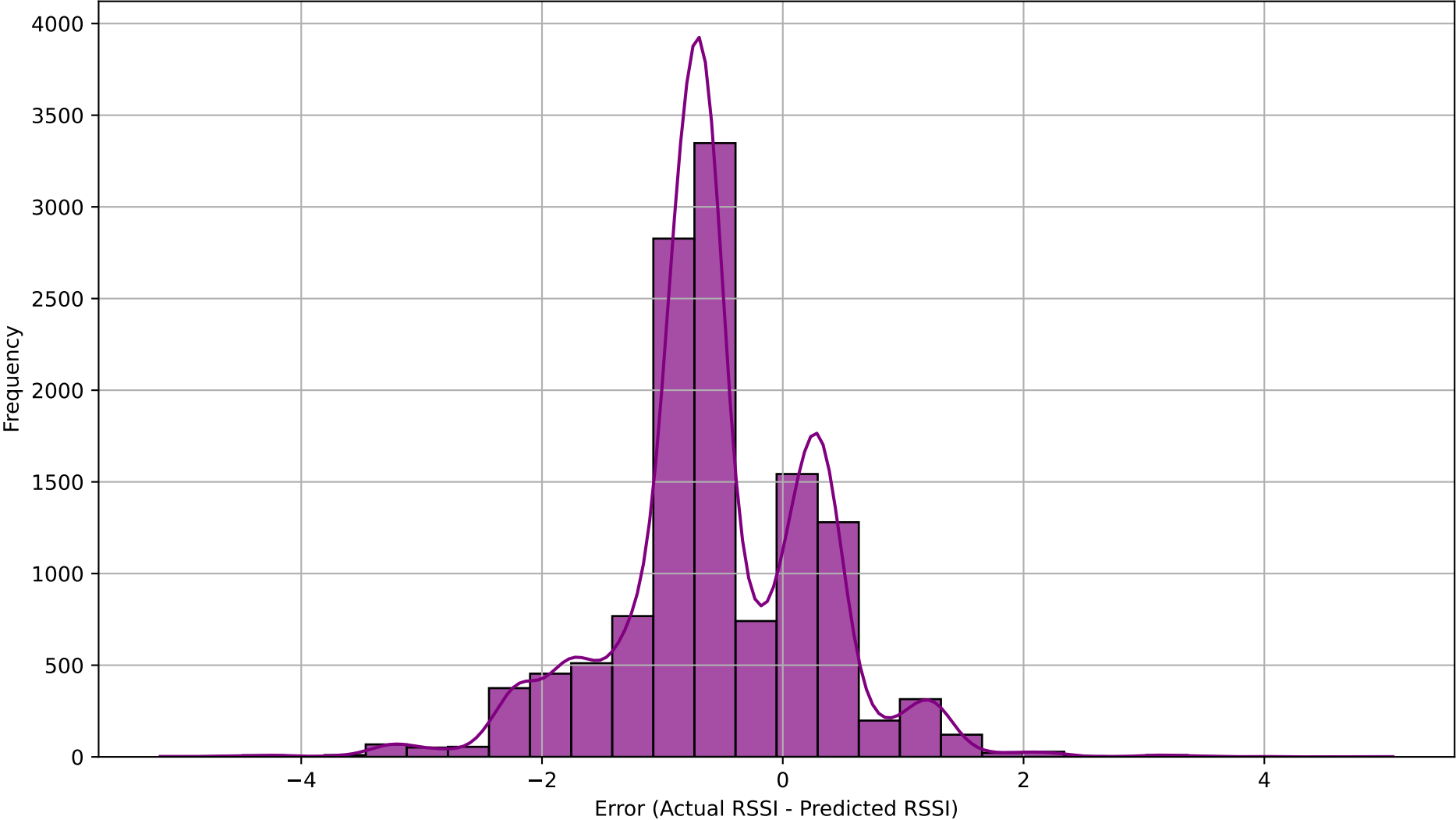
Actual vs Predicted RSSI (First Step)



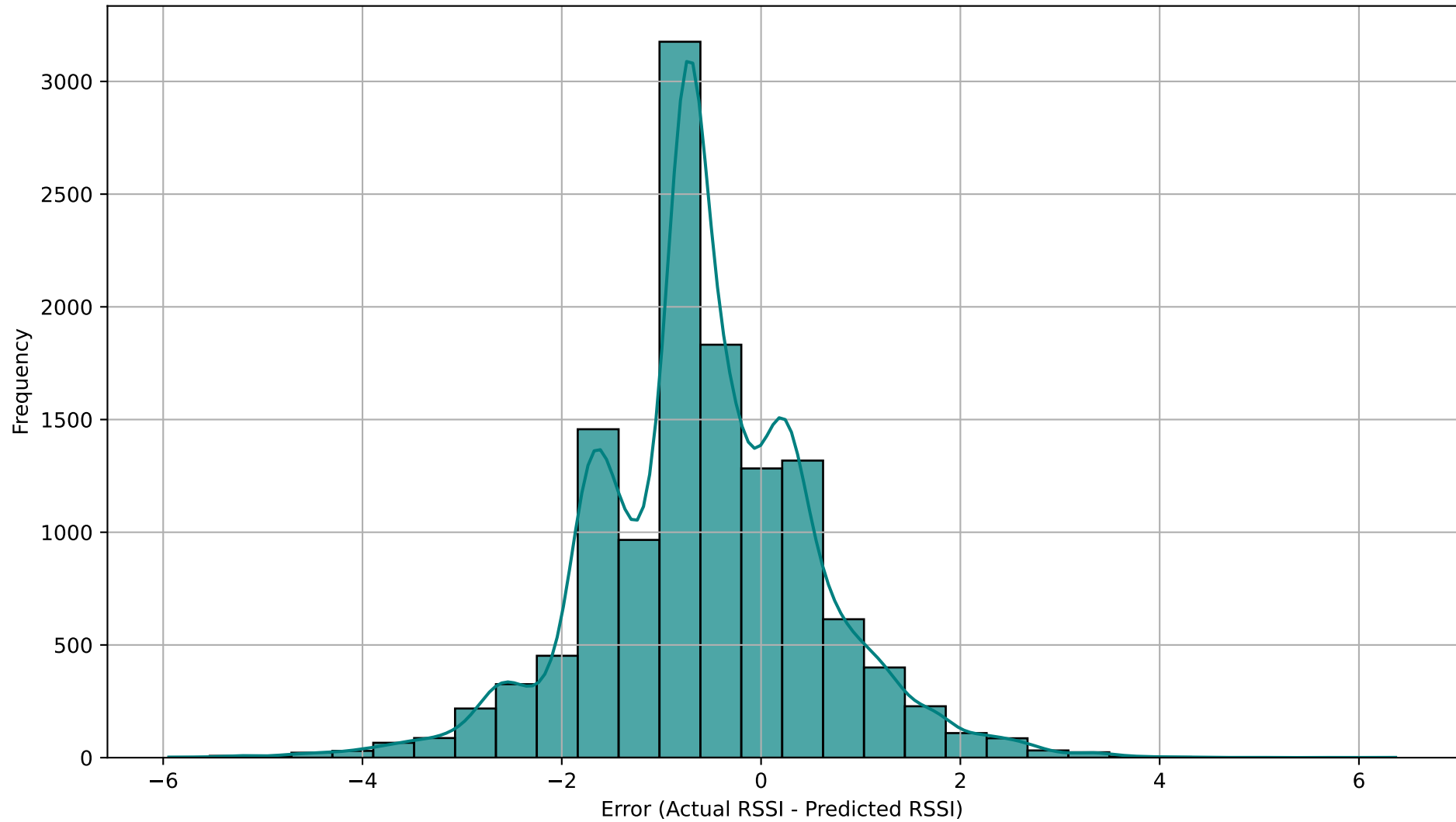
Actual vs Predicted RSSI (Last Step - step 5)



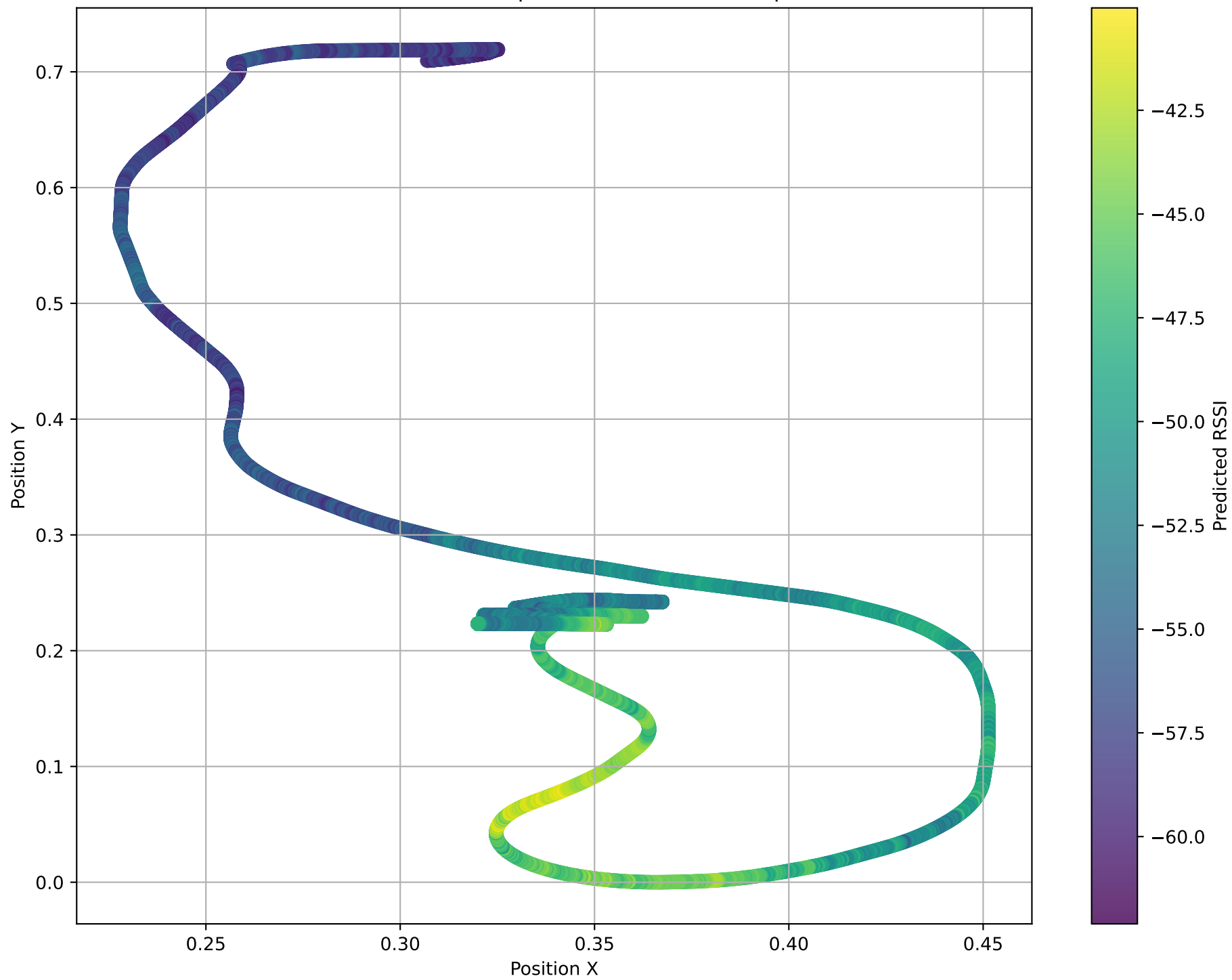
Distribution of Prediction Errors (First Step)



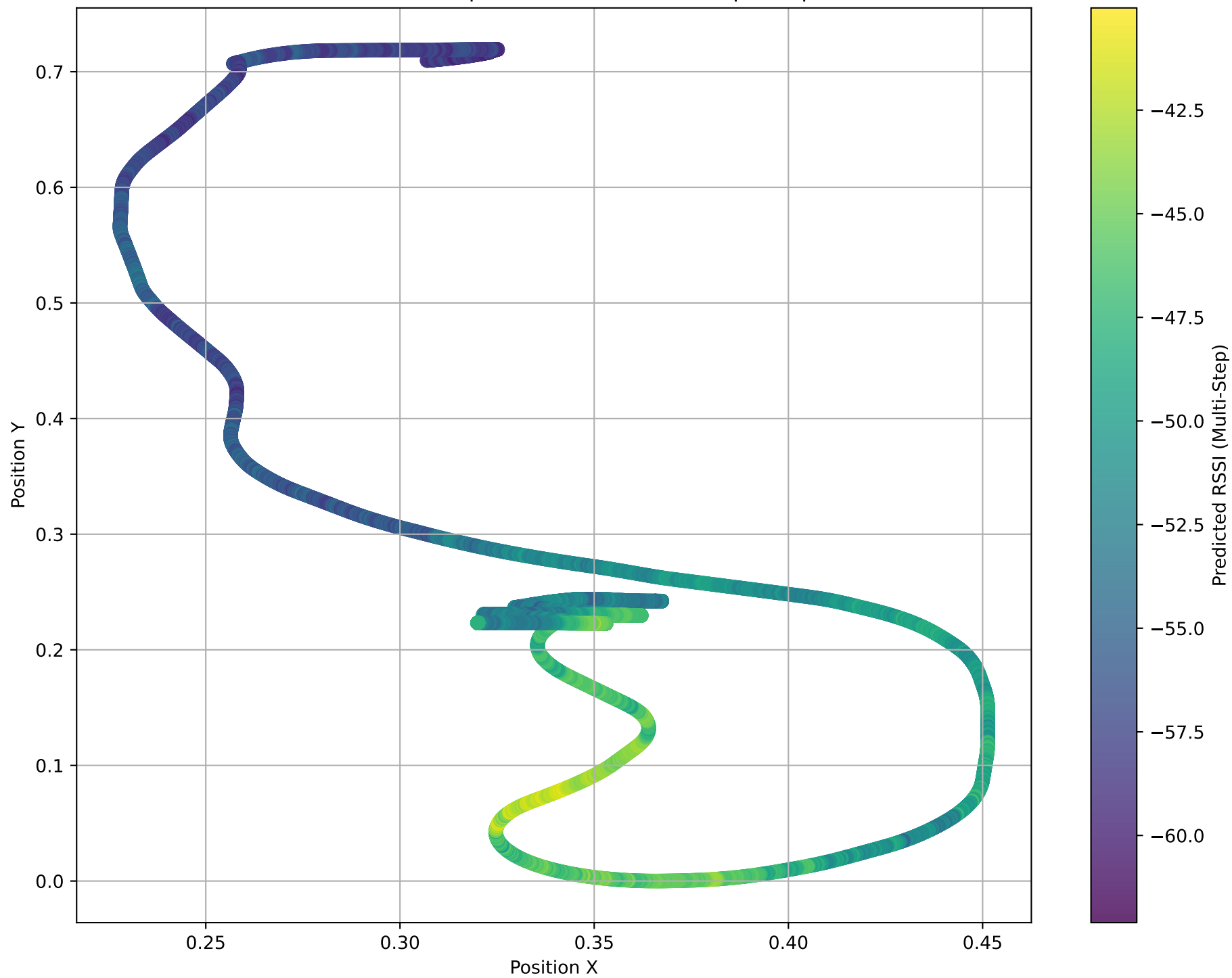
Distribution of Prediction Errors (Last Step - step 5)



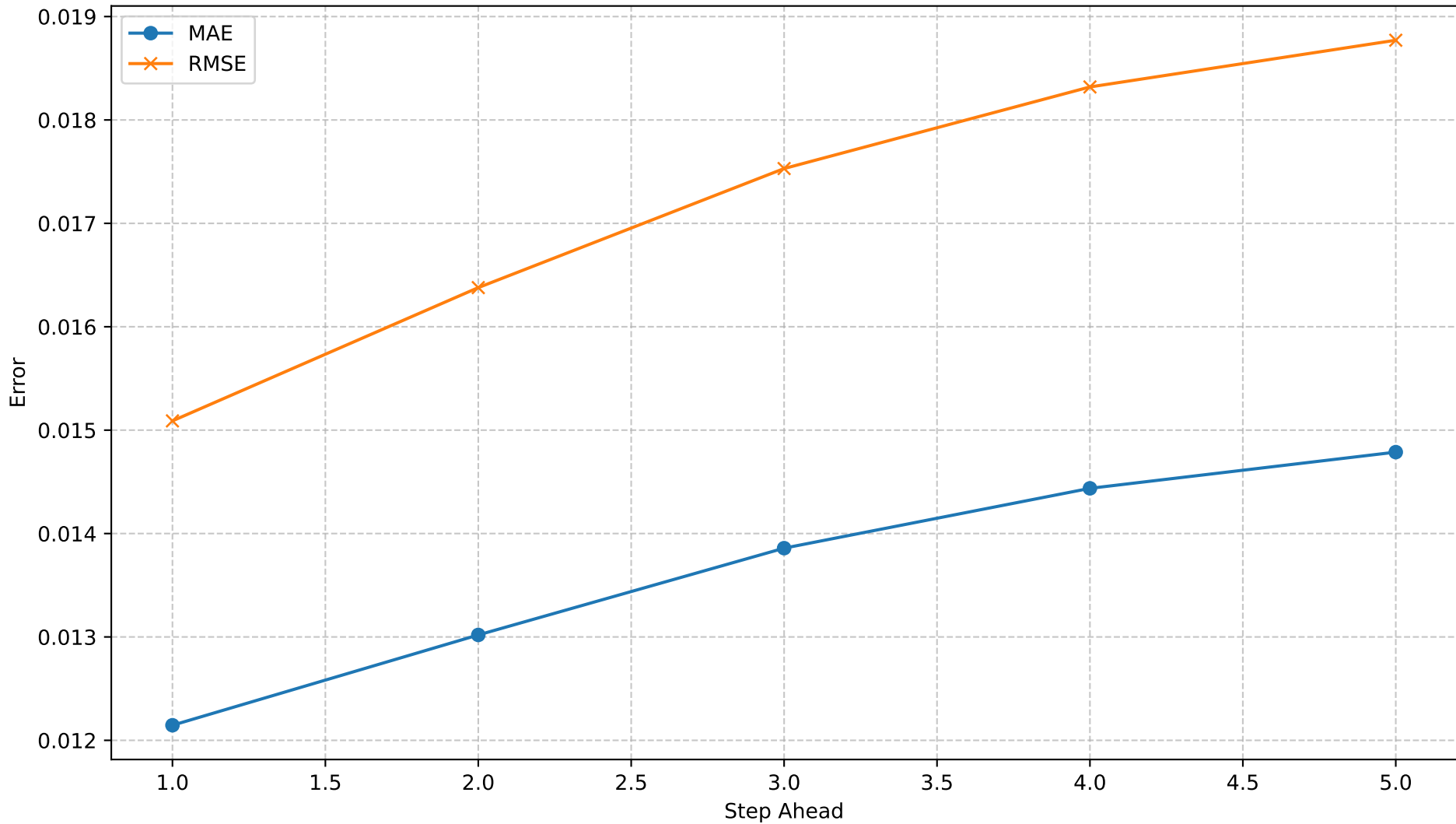
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 5)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 5

MAE / RMSE (dB)

- 1º passo : 0.01 / 0.02
- Mediana : 0.01 / 0.02
- 5º passo : 0.01 / 0.02

Crescimento do erro: 21.7% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 5 (0.01), 4 (0.01), 3 (0.01)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.